

SHIV KUMAR

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EDUCATION

Vellore Institute of Technology

Integrated M.Tech in Computer Science and Engineering (Computational and Data Science)

2022 – 2027

Bhopal, India

EXPERIENCE

Software Development Intern

Oct 2025 – Dec 2025

Imperative Code

Jaipur, India

- Designed and normalized relational schemas in MySQL for enterprise-scale management systems deployed across schools and coaching centers, ensuring scalable and maintainable data architectures.
- Performed exploratory data analysis and optimized complex SQL queries by interpreting database performance metrics, improving average query execution speed by ~30%.
- Collaborated with cross-functional development teams to implement reusable and well-documented backend components, improving operational workflow efficiency by ~25%.
- Prepared structured technical documentation and presented optimization findings, methodologies, and system performance insights to development teams and project stakeholders.

PROJECTS

Autonomous Multi-Agent AI Travel Orchestrator | Python, OpenAI API, ChromaDB, Agentic Workflows, AWS 2026

- Developed a multi-agent Generative AI system leveraging OpenAI API and Large Language Models (LLMs) to process unstructured travel requests and generate context-aware itineraries across 100+ simulated user scenarios.
- Integrated ChromaDB to support semantic retrieval and contextual grounding, reducing search latency by 40% while improving response consistency and context retention.
- Built a scalable FastAPI backend with asynchronous agent communication and secure API handling practices, enabling efficient processing of 100+ concurrent requests on AWS EC2.

AI-Powered Content Summarization & RAG Pipeline | Python, OpenAI API, ChromaDB, FastAPI 2026

- Implemented a Retrieval-Augmented Generation (RAG) pipeline utilizing OpenAI API and modern LLMs to automate document summarization workflows, reducing manual review effort by over 70%.
- Engineered semantic retrieval workflows using embeddings and ChromaDB to extract relevant context from 2,000+ documents while improving retrieval relevance and reducing hallucinated responses.
- Deployed the summarization engine as a modular FastAPI service with reusable API components and structured documentation to support scalable AI experimentation workflows.

Generative AI Interview Evaluation System | Python, OpenAI API, NLP, ChromaDB, FastAPI 2026

- Developed an AI-powered interview evaluation platform leveraging OpenAI API and NLP techniques to analyze conversational patterns across 500+ simulated interview transcripts.
- Implemented a Retrieval-Augmented Generation (RAG) architecture using ChromaDB to evaluate candidate responses against predefined technical criteria with strong evaluation consistency.
- Designed a FastAPI-based ingestion pipeline to process streaming textual data and support rapid prototyping, accelerating candidate screening workflows by 60%.

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, PySpark

Data Science & Machine Learning: Data Analysis, Predictive Modeling, Feature Engineering, Embeddings, Model Evaluation, Linear Regression, Logistic Regression, Decision Trees, Random Forest, XGBoost, K-Nearest Neighbors (KNN), Naive Bayes, K-Means Clustering, DBSCAN, Principal Component Analysis (PCA)

Generative AI & NLP: Natural Language Processing (NLP), Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Agentic AI, LangChain, OpenAI API, Vector Databases, Semantic Search, Fine-Tuning LLMs

Deep Learning: Artificial Neural Networks (ANN), Convolutional Neural Networks (CNN), Transformers

Cloud & API Development: Amazon Web Services (AWS), FastAPI

Libraries & Frameworks: Pandas, NumPy, Scikit-learn, PyTorch

Core Computer Science: Data Structures and Algorithms, DBMS, Operating Systems

CERTIFICATIONS

AWS Certified Solutions Architect - Associate (SAA-C03) | AWS

April 2026